

Raju Roopani

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PROFESSIONAL SUMMARY

Principal Engineer with 14+ years of distributed systems and platform engineering at planetary scale; the last 3+ years focused on real-time conversational AI and agentic systems. At Microsoft Teams architected the Meetings AI layer - real-time RAG over meeting context (transcript, calendar, task systems), context-window optimization, and streaming LLM integration delivering sub-200ms AI assistance to millions of daily participants. Full-stack delivery from React/TypeScript frontend to distributed Python/Java backend, with deep ML/Eng collaboration patterns. Looking to apply real-time RAG and conversational AI product engineering depth to a focused conversational AI company.

CORE COMPETENCIES

Conversational AI

Real-Time RAG

Full-Stack (React + Python)

No-Code-Adjacent Platforms

LLM Integration & Streaming UI

Performance Optimization

ML / Eng Collaboration

Production AI at Scale

WORK EXPERIENCE

Microsoft

Jul 2018 - Present

Principal Software Engineer | AI Tech Lead - Teams Platform

Redmond, WA

- Real-Time Conversational AI - Teams Meetings AI:** Built the real-time conversational AI layer for Teams meetings - hybrid **RAG retrieval pipeline** over meeting context (transcript, calendar, task systems), context-window optimization, and streaming UI integration with the LLM tool-use loop. **Sub-200ms perceived latency** for AI assistance to millions of daily participants.
- Full-Stack No-Code-Adjacent Platform - Modern Project Online:** Built collaborative no-code-style platform interfaces for non-technical PMs - React/TypeScript with Fluent UI Data Grid optimization, GraphQL APIs, dynamic UI configuration. **30% page load reduction** via virtualization, memoization, and batch update strategies for millions of enterprise users. Visual configuration UI for non-engineers - same product shape as no-code editors.
- Agent SDK for Less-Technical Developers:** Designed and shipped the Teams Agent SDK - manifest-based declarative agent configuration, opinionated defaults for HITL, retrieval, and error handling. Adopted by 3rd-party developers (varying technical depth). **3rd-party agent reliability +3x; developer support tickets -60%**. Same shape of problem as platforms for non-technical users.
- Multi-Agent Orchestration Runtime:** Architected and delivered a real-time distributed multi-agent orchestration runtime supporting Agent-to-Agent (A2A) protocol, Model Context Protocol (MCP), HITL, partial-failure isolation - serving **300M+ daily active users**.
- Latency, Cost, & Quality Optimization:** Reduced p95 agent runtime latency from 320ms to 85ms via event-driven coordination, retrieval caching, connection pooling, and tiered model routing. Cut cost per request by 40%.
- ML / Eng Collaboration:** Built integration patterns - eval pipelines, observability instrumentation, progressive rollout - that let AI Research ship model improvements without product regression. Cross-functional TL across Engineering, Product, Design, and AI Research.

Envestnet Yodlee Infotech

Apr 2016 - Jul 2018

Member of Technical Staff - Platform Engineering

Bengaluru, India

- High-Throughput REST APIs:** Designed and developed REST APIs aggregating financial data across banks and third-party providers, processing **10M+ transactions per day** on event-driven Kafka pipelines using Java Spring Boot.
- Framework-Agnostic Integration Platform:** Built integrations bridging hundreds of heterogeneous external systems (100+ bank APIs with varying schemas and custom auth) into a unified runtime.
- OAuth 2.0 + PSD2 Compliance:** Architected and implemented OAuth 2.0 client framework in Java Spring MVC and Hibernate, enabling PSD2-compliant integration with EU Open Banking APIs across multiple European markets.

Software Engineer - Build & Automation Tooling

Bengaluru, India

- **Build Systems & Automation Tooling:** Designed and built automation tooling for Physical IP modeling and validation flows - implementing complex verification algorithms, efficient build pipelines, and reusable tooling frameworks. Reduced manual verification effort by ~60% and accelerated hardware design-to-tape-out cycles.

EDUCATION

B.E. - Electronics & Communications Engineering · [Osmania University](#)

2007 - 2011

SKILLS

Conversational AI: Real-Time RAG, Context Engineering, LLM Integration, Streaming UI, Conversation Workflows, Agent Orchestration, A2A Protocol, MCP, HITL

LLM Frameworks: AutoGen, LangChain (familiarity), LlamaIndex (familiarity), Azure OpenAI, OpenAI API, Anthropic Claude API

Languages: Python (AI integrations: AutoGen, Azure OpenAI SDK, Anthropic SDK, OpenAI API), TypeScript, C#/.NET, Java (Spring Boot), Node.js

Frontend: React, TypeScript, Fluent UI, GraphQL, Performance Optimization (virtualization, memoization, batch updates), Streaming UI

Backend & APIs: C#/.NET, Java Spring Boot, REST, GraphQL, gRPC, WebSockets, Microservices, Event-Driven Architecture

Distributed Systems: Microservices, Event-Driven Architecture, Fault Tolerance, Horizontal Scalability, Idempotency, Partial-Failure Isolation

Cloud & Infra: Azure (Service Bus, Functions, Cosmos DB, OpenAI), Docker **Messaging:** Apache Kafka, Azure Service Bus

Databases: Cosmos DB, MySQL, MongoDB, Oracle (relational + document patterns transferable to Postgres / ClickHouse / Elasticsearch)